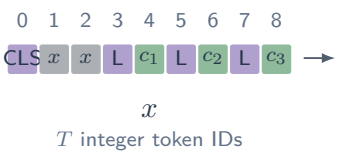
$$q = f_{text}(H_{CLS,:}), \quad L = f_{label}(H[P_L, :]), \quad s = qL^{\mathsf{T}}, \quad p = \sigma(s)$$

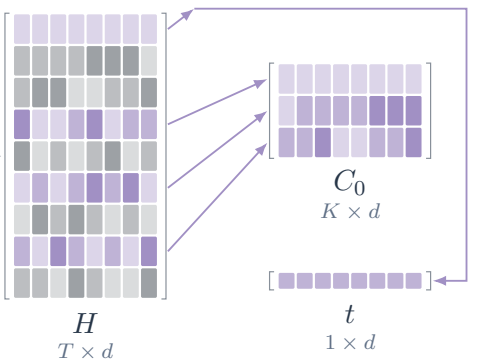
### 1. Token layout




### 2. GLiClass

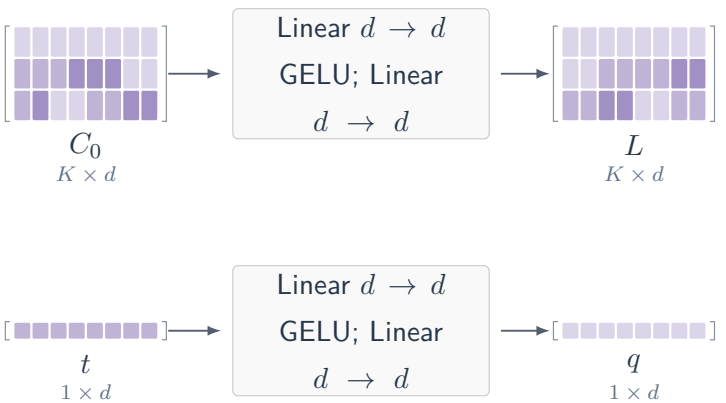
DeBERTa-v3-small
6 bidirectional
blocks; d=768
uni-encoder /
simple scorer

### 3. Contextual states and indices

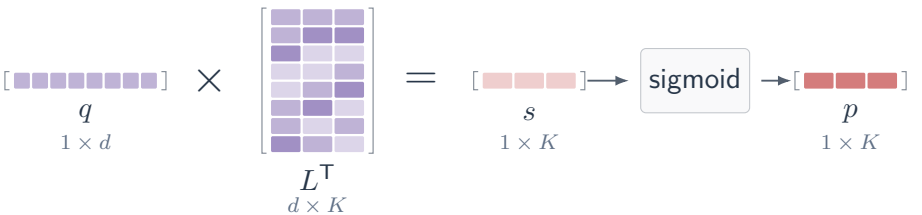



Joint bidirectional input
read CLS + label markers
no feature L2;
no temperature

### 4. Distinct text and label projectors




### 5. Text-label dot product




| | |
|---|---|
| **Axes** | $d = 768$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes. |
| **Objects** | Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes. |
| **Mechanism** | CLS and label markers see the joint input. Independent d-to-d-to-d projectors give a text/label dot product; per-label sigmoid does not imply independent pairs.<br>Named multi-label loss branch is BCE; the old checkpoint training history is not recovered. The selected config has no feature L2 or temperature scale. |